EE-M-131G
July 24, 1979
SUPERSEDING
Fed. Spec. EE-M-131F
June 27, 1975

FEDERAL SPECIFICATION

MAYONNAISE AND SALAD DRESSING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for mayonnaise and salad dressing for use by all Federal agencies. Types IV through X shall conform to the physical requirements of 3.3.1, but not the analytical requirements of 3.3.2.

1.2 Classification.

1.2.1 Types, styles, and classes. Mayonnaise and salad dressings covered by this specification shall be of the following types, styles, and classes, as specified (see 6.1):

Type I - Mayonnaise
Type II - Salad dressing
Type III - French dressing

Style A - Emulsified
  Color 1 - Orange-red (with added tomato product)
  Color 2 - Other than orange-red (with or without tomato product)

Style B - Separating
  Color 1 - Orange-red (with added tomato product)
  Color 2 - Other than orange-red (with or without tomato product)

Classes. Type I mayonnaise, type II salad dressing and type III French dressing covered by this specification shall be of the following classes, as specified (see 6.1):

Class 1 - Cottonseed oil, corn oil or specially processed soybean oil separately or in combination with each other (see 3.1.2 for analytical requirements as applicable)

Class 2 - Soybean oil or other edible vegetable oils in common domestic use (except olive oil) not specified (see 3.1.2 for analytical requirements as applicable)

FSC 8950

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Type IV	-	Thousand Island Dressing	
Type V	-	Blue Cheese dressing	
Type VI	-	Italian dressing	
Type VII	-	Cole Slaw dressing	See 6.1b
Type VIII	-	French dressing	
Type IX	-	Tartar sauce	
Type X	-	Green Goddess dressing	

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-G-460	-	Glass Containers, Filled and Closed; Packaging and Packing Of
PPP-I-350	-	Individual Servings of Subsistence; Packaging and Packing Of

Federal Standard:

FED-STD-123	-	Marking For Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston, Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in the agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items.
MIL-L-35078	-	Load and Unit Preparation of Nonperishable Subsistence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Definitions and Standards for Dressings for Foods, Mayonnaise, French Dressing and Salad Dressing

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U. S. Department of Agriculture

Regulations Governing the Grading and Inspection of Egg Products

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, Poultry Division, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U. S. Department of Agriculture, Washington, DC 20250.)

Egg Products Inspection Handbook

FSQS - PY Instruction number 910 (Egg products - 1)

(Application for copies should be addressed to the nearest office of the Poultry Grading Branch of the Poultry and Dairy Quality Division, FSQS, USDA, as listed below:

469 Hospital Drive, Hospital Plaza Building, Gastonia, NC 28052
1 National Resources Drive, Room 110, Little Rock, AR 72105
210 Walnut Street, Room 777, Des Moines IA 50309
Federal Building, Room 219, 410 North San Joaquin, Stockton, CA 95202

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Oil Chemists Society, Inc.

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 503 South 6th Street, Champaign, IL 61820.)

American Public Health Association, Inc.

Compendium of Methods for the Microbiological Examination of Foods of the American Public Health Association, Inc.

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W., Washington, DC 20036.)

U. S. Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 46 Park Avenue, New York, NY 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. All component materials shall be clean, sound, wholesome, of good flavor, aroma, and of food grade quality. They shall be free from insect infestation and other foreign matter. In addition, the materials for types I, II and III shall comply with the requirements in 3.1.2. through 3.3.1.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.1.1 Bid sample approval (Military agencies). When specified (see 6.1), the bidder shall submit with his bid six one half pint (237 ml) factory filled and sealed glass bottle samples representative of each type product (type IV through type X) which he proposes to furnish if awarded a contract. The samples shall be submitted to the contracting officer with the bid or otherwise furnished prior to bid opening. Three of the six samples shall be evaluated by a trained panel for flavor, odor, texture and appearance requirements of 3.3.1. The remaining samples submitted by a successful bidder shall be used by the contracting officer as standard reference samples to determine compliance with the procured item. The approval of any bid sample for flavor, odor, texture and appearance shall not constitute approval of the sample as meeting any other requirements of this specification.

3.1.2 Vegetable oils (see 1.2.1, classes 1 and 2). All vegetable oils shall be refined and deodorized in accordance with good commercial practice and at time of use, shall have a bland flavor and odor, free from any objectionable off-flavors and off odors. Heavy metal scavengers may be added to improve the keeping quality and use performance of the oils. The use of such scavengers shall be in accordance with 3.4. In addition, the oils shall meet the following analytical requirements.

Free fatty acids, not more than	0.05 percent expressed as oleic acid
Peroxide value, not more than	1.0 milli-equivalent per 1000 g.
Cold test, not less than	10 hours 1/
Linolenic acid, 2/, not more than . . .	0.50 percent by weight
Linolenic acid, 3/, not more than . . .	3 percent by weight

1/ Applicable to cottonseed oil, corn oil and specially processed soybean oil.

2/ Applicable only for class 1, all types. When corn oil is used exclusively, a maximum of 0.90 percent linolenic acid by weight is permitted. When the blend consists of a high level of specially processed soybean oil, the maximum linolenic acid content shall be based on a calculated and proportioned basis, (see also 4.5.2.1.10 for tolerance).

3/ Applicable only for class 1, all types when specially processed soybean oil is used as the sole oil component.

3.1.3 Egg products. Liquid egg yolks, frozen egg yolks, dried egg yolks, liquid whole eggs, frozen whole eggs, dried whole eggs or any one or more of the foregoing with liquid egg white or frozen egg white may be used in the preparation of the product provided that the standard plate count for bacteria for such egg products does not exceed 100,000 per gram. Egg product shall have been processed in an official plant pursuant to the Regulations Governing the Inspection of Eggs and Egg Products, shall be eligible to be identified in accordance with section 59.412 thereof and are accompanied by U.S. Department of Agriculture certificates which attest to meeting the standard plate count

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

requirement and that the products are salmonellae negative, using the sampling plan pattern described under "Surveillance Sampling" Section 8 of USDA AMS, PY - Instruction No. 910, Egg Products Inspection Handbook. Frozen eggs shall be held at 0°F (-18°C) or below until thawed immediately before use. Fresh, broken whole eggs shall be held at a temperature not higher than 40°F (4°C) and shall be used within 72 hours from time of breaking. In addition, when frozen whole eggs are used in preparation of the product, they shall not have been stored longer than 9 months. When frozen egg whites or frozen yolks (sugar or salt added) are used in preparation of the product, they shall not have been stored longer than 12 months.

3.2 Other ingredients. Only ingredients (spices, starches, stabilizers, chelating agents, sugars, and other ingredients) which conform to those permitted in the Definitions and Standards for Dressings for Foods, Mayonnaise, French Dressing and Salad Dressing, under the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder, shall be used in the preparation of these products.

3.2.1 Carbon dioxide. Carbon dioxide shall meet the requirements of the Pharmacopoeia of the United States of America.

3.3 Description of types. The products shall be composed of the ingredients permitted in the Definitions and Standards of Dressings for Food, Mayonnaise, French Dressing and Salad Dressing, under the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.3.1 Physical requirements. The physical requirements for types I, II, and III shall be as follows:

Types I and II

Flavor and color:	A pleasing flavor and color typical of good commercial product.
Body:	Semisolid and does not flow from its inverted container when product is below a temperature of 90°F (32°C).
Texture:	Uniformly smooth and free from lumps.
Emulsion stability:	Exhibits no separation of oil or water phases after 56 hours when tested according to the emulsion stability test described in 4.5.2.2.

Type III, style A, emulsified, color 1

Flavor:	A pleasing flavor and color typical of good commercial product.
Color:	Orange-red derived from added tomato product.
Body:	The emulsified product shall have a thick pourable consistency when it is at a temperature of 40°F -50°F (4°C -10°C).
Texture:	Uniformly smooth and free from lumps.
Emulsion stability:	Exhibits no separation of oil and water phases after 168 hours when tested by the emulsion stability test described in 4.5.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Type III, style A, emulsified, color 2

Product shall comply with all the requirements of type III, style A, except color may be other than orange-red. Product may contain added tomato product.

Type III, style B, separating, color 1

Flavor: A pleasing flavor typical of good commercial product.

Color: A temporary orange-red color, derived from added tomato product, immediately after thoroughly shaking the product in its container.

Body: The product after thorough shaking in its container, shall be characteristically thick but readily pourable at 40°F -50°F, (4°C -10°C). The temporary emulsion shall not separate visibly during pouring immediately after shaking.

Texture: Uniformly smooth and free from lumps.

Type III, style B, separating, color 2

Product shall comply with all the requirements of type III, style B, except color may be other than orange-red immediately after thoroughly shaking the product in its container. Product may contain added tomato product.

Types IV through X

These types of dressings represent high grade commercial brand name products with excellent product quality characteristics and shall be formulated from oil meeting the requirements of 3.1.2, as applicable.

Type IV - Thousand Island dressing. Thousand Island dressing shall be made using a high grade commercial quality mayonnaise product as a base to which has been incorporated pickle relish. Optional flavoring ingredients may be added, such as pimento, onion, garlic, chili sauce and chopped cooked egg yolks. The product shall have a tangy, zestful, spicy flavor and aroma. The dressing shall have a thick, creamy, pourable consistency and shall be free from separation and gas formation.

Type V - Blue Cheese. Blue Cheese dressing shall be made using a high grade commercial quality mayonnaise product as a base to which has been incorporated Blue Cheese or Roquefort or a blend of both. The cheese may be in the form of crumbles, granulated cheese or discrete pieces. Optional ingredients may be Worcestershire sauce, spices, sweeteners, salt and pepper. The product shall contain, in addition, garlic as a supplementary seasoning. Blue Cheese dressing shall have a thick, creamy, pourable consistency and shall be free from separation and gas formation. The product shall have an agreeably pungent, sharp, spicy, peppery, piquant flavor and odor.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:43Z
Check the source to verify that this is the currentversion before use.

Type VI - Italian dressing. Italian dressing (separating type) shall consist of an oil phase and a vinegar brine phase (i.e., using separate oil and vinegar/brine phases), to which has been incorporated lemon juice, finely minced garlic cloves, onion and red peppers. Optional ingredients may include finely broken up, dried oregano leaves, sugar, paprika, dried mustard, Worcestershire sauce, finely chopped parsley, grated Parmesan cheese or other suitable spices and spice flavoring used in commercial production. The addition of a temporary emulsifier is permitted. The product shall have a tangy, zestful, spicy flavor and aroma. In addition, the dressing shall meet the body characteristics for type III, style B.

Type VII - Cole slaw dressing. Cole slaw dressing shall be made from ingredients permitted in the Standards of Identity for Mayonnaise or Salad Dressing. If desired, mayonnaise or salad dressing may be used as a base material with incorporation of additional sweetening, acidifying and spicing ingredients. The product may vary in color from a pale yellow to that of a deeper egg yolk shade characteristic of mayonnaise. The dressing shall be thick, creamy and smooth and shall be homogeneous in appearance, free from gumminess and show no separation of oil or watery phases. The flavor and aroma shall be sweet, tart, zestfully savory and mildly spicy.

Type VIII - French dressing. French dressing shall be made in accordance with requirements promulgated in the Standards of Identity for this product using appropriate ingredients permitted herein. The finished product shall be the emulsified type and it shall have an orange-red color. French dressing shall have a creamy thick, pourable consistency at 40° to 50°F (4° to 10°C) and shall be homogeneous in appearance, free from separation and gas formation. The product shall have a tangy, zestful, spicy flavor and aroma, free from stale oxidized, oily, beany, rancid, painty, and other undesirable flavors and aromas.

Type IX - Tartar sauce. Tartar sauce shall be made using a high grade commercial quality mayonnaise product as a base to which has been incorporated suitable processed dill or sour pickles, or a commercial type of stabilizer and pickle relish containing spices commonly used for tartar sauce. Other added ingredients may consist of capers, onions, cucumbers, or peppers in amounts and in particle size normally used commercially. The product shall have a thick creamy consistency suitable for a sandwich spread. There shall be no separation of the oil or water phases. The incorporated vegetable pieces shall be firm and crunchy. The flavor shall be pleasantly tart with overtones of the characteristic flavors contributed by the various components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Type X - Green Goddess dressing. Green Goddess dressing shall be made using a high grade commercial quality mayonnaise product as a base to which has been incorporated minced anchovies or anchovy paste, chopped chives, and spice flavor commonly used in commercial formulations. Other ingredients may consist of milk products, celery seed, minced parsley sprigs, tarragon, vinegar, and lemon juice. The blended product shall have a creamy, thick, pourable consistency at 40° to 50°F (4° to 10°C), shall be homogeneous in appearance, and free from separation, gas formation, or air voids. The product shall be of the sweet/sour flavor type with a subtle anchovy or characteristic fish flavor background and without being objectionably salty. The added chopped or minced components, including celery seed when used, shall be uniformly distributed throughout the product. The color shall be a pale green.

3.3.1.1 Flavor. All types, styles, and classes shall be flavorsome and highly palatable, and when freshly packaged, free from objectionable flavors and odors such as oxidized, tallowy, rancid, beany, painty, yeasty and starchy.

3.3.1.1.1 Foreign material. All types shall be free from foreign material, such as rust, dirt, grease, wood particles, paint, lacquer, insect parts, glass, hair, metal, and paper.

3.3.2 Analytical requirements. Analytical requirements shall comply with the following:

Analytical requirements

	Types I	II	III
1. Linolenic acid percent not more than 1/ 2/ 3/ (applicable to class 1 only)			
Cottonseed oil	0.50	0.50	0.50
Corn oil	0.90	0.90	0.90
Specially processed soybean oil	3.00	3.00	3.00
2. Total fat, percent, not less than 4/	80.00	35.00	35.00
3. Acidity, percent of moisture phase, not less than	2.50	2.25	2.75
4. Egg yolk, percent not less than 5/	7.00	5.00	---
5. Total sugars (reported as invert) percent by weight, not less than	---	8.00	12.00
6. Copper, not more than ppm	2 ppm	2 ppm	2 ppm

1/ Determined on the oil separated from the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

2/ Linolenic acid values given above refer to the finished product when cottonseed oil, corn oil or specially processed soybean oil is used separately. When the oils are used in a blend of 2 or more, the larger value (3.00) will be acceptable.

3/ For test tolerance see 4.5.2.1.10.

4/ Minimal fat derived from vegetable oil shall meet the Standards of Identity for the products involved.

5/ Shall test or be standardized to contain 43 percent egg yolk solids.

3.3.3 Time of shipment. Unless otherwise specified (see 6.1), the finished product shall be shipped within ten (10) days after the date of manufacturing and packaging of the finished item.

3.4 Federal Food, Drug, and Cosmetic Act All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.5 Plant qualification. The product shall be prepared in accordance with part 110 of the Code of Federal Regulations, Title 21, Human Foods; Current Good Manufacturing, processing, packing or holding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification conditions. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.1 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.3 Acceptance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U. S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. For applicable criteria governing condition inspection of containers for military agencies see 4.3.4.

4.3.2 Sampling for inspection. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3.3 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable document.

4.3.3.1 Examination of ingredients. Conformance of ingredients to the requirements of 3.1.2 and 3.2 as concerns identity (kind, U.S. Pharmacopoeia compliance) shall be ascertained by a review of labels, invoices or other valid documents. In addition, each ingredient, except carbon dioxide, shall be examined organoleptically, as necessary, to determine conformance to the condition requirements of 3.1. The sample unit for examination shall be approximately one pound (.45 kg) of ingredient (except for minor additives such as spices, stabilizers and chelating agents, for which the sample unit shall be two ounces (57 g) derived from a primary container. Five primary containers shall be sampled, or all containers if less than five form a lot. Any nonconformance to a condition or identity requirement shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved quantity of finished product.

4.3.3.2 Examination of egg products. Compliance of these products with the requirements of 3.1.3 as concerns identity, grading and inspection shall be ascertained by examination for presence of USDA Inspection Shield and USDA certificate attesting to salmonella negative product and compliance with the standard plate count limitation. In addition, examination shall be made to determine compliance with the requirements concerning storage or holding temperature, time intervals and age. Such examinations shall be determined on the first lot and on subsequent lots as deemed necessary by the cognizant Government inspection activity. Use of nonconforming egg products shall be cause for rejection of the involved quantity of finished product.

4.3.3.3 Testing of components. Testing of components described in this specification shall be as shown in table I. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE I. Testing of components 1/ (see 4.3.3)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt para
Closure for glass containers	One closure	Closures	S-1	Tinplate	Nearest 0.01 lb/ base box (0.24 g/m2)	5.1.1.2 4.5.2.4
Oil, vegetable	One quart composite 2/ (945 ml)	Pounds (kg)	S-1	Free fatty acids	Nearest 0.01 percent as oleic acid	3.1.1 4.5.2.1.1
				Linolenic acid (see footnotes 1/ and 2/ to 3.1.2	Nearest 0.1 percent	4.5.2.1.10
				Peroxide value	Nearest 0.1 meq/kg of oil	4.5.2.1.2
				Cold test	Pass or fail	4.5.2.1.3

1/ Testing of components shall be applicable to pertinent material related to types I, II and III products only.

2/ Composite sample shall be derived from the number of containers indicated by the inspection level.

4.3.4 Examination of end item. Classification of defects found during examination shall be in accordance with tables III through VI and 4.3.4.2 through 4.3.4.7 and shall be limited to types I, II and III products. (For types IV through X see 4.3.4.1). The examination criteria for the above mentioned tables shall be as shown in table II. Acceptable quality levels (AQLs) shall be expressed in defects per hundred units for tables III and V and in terms of percent defective for tables IV and VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE II. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor	Total
III	I	One filled and closed glass container	Glass containers	0.25	1.5	-	4.0
IV	S-3	One filled primary container	Primary containers	-	-	2.5	-
V	S-2	Contents of one primary container	Primary containers	1/	1.5	-	-
VI	S-2	One labeled glass container	Glass containers	-	-	2.5	-

1/ The presence of any major A defect shall be cause for rejection of the involved lot of finished product.

TABLE III. External examination of glass containers

Category			Defect
Major A	Major B	Minor	
	151		Not size specified
	152		Not either one of types of closure specified
	153		Shrink or cellulose band or pressure-sensitive tape, when required, missing, misplaced or not overlapping as specified, or when applicable, not size specified
101			Leaker
102			Cracked
	154		Chipped or thin spot
		201	Not clean
	155		Pitted rust on closure 1/
	156		Closure insecure or not hermetically sealed 2/
		202	Screwcap closure cross threaded (when applicable)
	157		Coatings on closure missing, incomplete or not as specified
	158		Inside liner for closure missing or not as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE III. External examination of glass containers (cont'd)

Category			Defect
Major A	Major B	Minor	
	159		Label missing
		203	Label not applied as specified
	160		Nomenclature on label missing, incorrect, or illegible
		204	Required label information other than nomenclature missing, incorrect, or illegible

1/ Rust shall not be considered pitted if it can be wiped off with a soft rag.

2/ Use industry accepted torque tester. Torque removal (inch pounds) shall be as indicated:

Closure size (mm)	Tightness (inch pounds)
38	15-23
53	21-32
70	28-42
86	35-51
110	45-65

TABLE IV. Examination of net weight or volume for can and glass containers 1/

Category	Defect
Minor	
201	Net weight or volume, as applicable, more than two percent under specified weight or volume

1/ Report results to nearest 1/2 ounce, (nearest gram,) (or nearest ml,) as applicable, for containers other than gallon (3.7 liter) jars. Report results for gallon (3.7 liter) jars to nearest 1 ounce, nearest gram, or nearest ml., as applicable. Lot shall be rejected if sample data indicate lot average net weight or volume, as applicable, are less than specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE V. Examination of product characteristics 1/ 2/

Category		Defect
Major A	Major B	
101		Presence of foreign material (e.g., rust, dirt, grease, wood particles, paint, lacquer, insect parts, glass, hair, metal, paper, etc.)
102		Evidence of objectionable flavors and odors (e.g., oxidized, tallowy, rancid, beany, painty, yeasty, starchy, etc.)
	151	Not type style, class or color as specified
	152	Not flavorsome or not highly palatable when freshly packed 3/
	153	Not semisolid or flows from its inverted container when product is below 90°F (32°C) (for types I and II), or lacks in thick viscosity (for type III)
	154	Does not flow from its container when product is at a temperature of 40°F (4°C) or above (for type III)
	155	Not uniformly smooth or free from lumps

1/ See footnote 1/ to table II.

2/ Examination limited to types I, II and III only.

3/ Slight surface discoloration of the product adjacent to the side seam shall not be classified as a defect. (Applicable only to canned product.)

TABLE VI. Examination of label adhesion (see 5.3.2.1.2 and 4.5.1) 1/

Category		Defect
Major B	Minor	
	201	Label can be moved by finger

1/ Applicable to overseas (level A) shipment only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.3.4.1 Examination of product characteristics for types IV through X. Examination of these product types shall be made to determine compliance with the product characteristics specified in 3.3.1, as applicable. The sample for examination shall be approximately 16 ounces (473 ml) derived from each of five primary containers or all containers if less than five form a lot. Nonconformance to one or more of the above referenced condition requirements for the specific type of product, or failure of the product to be characteristic of the type associated with high grade, commercial brands or nonconformance to 3.3.1, 3.3.1.1 or 3.3.1.1.1, shall be cause for rejection of the involved lot of finished product.

4.3.4.2 Examination of cans. Examination of cans shall be made in accordance with the applicable criteria of PPP-C-29. In addition, the use of cans which do not have double enameled interiors and inside side seam stripes shall be cause for rejection of the involved lot of finished product. Such conformance shall be ascertained by a review of pertinent labels, invoices or other valid documents.

4.3.4.2.1 Examination of can vacuum. Examination for can vacuum shall be in accordance with PPP-C-29.

4.3.4.3 Examination of can label. Examination shall be in accordance with the criteria of MIL-L-1497.

4.3.4.4 Examination of individual servings and packaging, packing and marking of intermediate containers and shipping containers of individual servings. Examinations shall be in accordance with the applicable criteria of PPP-I-350. In addition, the following defects shall be included in the examination of the intermediate carton or box and shipping container: Marking defects shall be classified as follows: Major: NSN, contract number, item description or date of pack missing, incorrect or illegible. Minor: Other required markings, missing, incorrect or illegible.

4.3.4.4.1 Examination of individual servings for net weight. Examination shall be in accordance with PPP-I-350.

4.3.4.5 Examination for packaging and packing of glass containers. Examination shall be in accordance with the applicable criteria contained in the appendix of PPP-G-460. Marking defects shall be classified as follows: Major: NSN, contract number, item description, or date of pack, missing, incorrect or illegible. Minor: Other required markings missing, incorrect or illegible. For level C pack, only the defects for workmanship and marking shall apply.

4.3.4.6 Examination for can packing and shipping container. Examination shall be in accordance with the applicable criteria of PPP-C-29. Marking defects shall be classified as follows: Major: NSN, contract number, item description or date of pack missing, incorrect or illegible. Minor: Other required markings, missing, incorrect or illegible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.3.4.7 Examination of unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.3.4.8 Examination of age of product. Examination shall be made at time of shipment of finished product to determine compliance with requirement of 3.3.3 as concerns age of product. Determination of compliance shall be ascertained by marked date of pack, codes or other valid means. Noncompliance to the above referenced requirement shall be cause for rejection of the involved lot.

4.4 Sampling procedure and acceptance criteria for testing of finished product. The finished product(s) shall be tested for linolenic acid, class 1 product, fat, acidity, egg yolk (types I and II), total sugars (as invert) (types II and III), copper and emulsion stability (types I and II and III, style A) in accordance with the requirements of 3.3.1 and 3.3.2. Procedures for testing shall be in accordance with 4.5. The sample for testing shall be an approximate one pound composite derived from the number of containers indicated by inspection level S-2, except that a sufficient number of cups, boats and pouches shall be taken to form the composite. In addition, the sample unit for the emulsion stability test shall be one primary container (see 4.5.2.2). Lot size shall be expressed in terms of primary containers. The results shall be reported as follows: Linolenic acid, fat, acidity, egg yolk and total sugars - to the nearest 0.01 percent; copper - nearest ppm; and emulsion stability - pass or fail. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

4.5 Examination and test methods. Examination and test procedures and controls which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the examination and test procedures set forth herein. In case of dispute as to examination of test results, the methods specified herein shall govern.

4.5.1 Examination of label adhesive for water resistance. The label adhesive shall be examined for water resistance as follows: Submerge glass container with label affixed into water at room temperature for four hours. Remove from water and remove excess water by blotting with paper towels. Press finger firmly against the label (in area where adhesive has been applied) and move finger still pressed against the label towards the bottom of the bottle. Displacement of the label indicates failure to meet the requirements.

4.5.2 Test procedures.

4.5.2.1 Chemical analyses. Chemical analyses shall be made in accordance with the methods specified below:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.5.2.1.1 Free fatty acid. Free fatty acid shall be determined in accordance with Official and Tentative Methods of the American Oil Chemists Society. Section C, Commercial Fats and Oils, Method Ca 5a-40.

4.5.2.1.2 Peroxide value. Peroxide value shall be determined in accordance with Official and Tentative Methods of the American Oil Chemists Society, Section C, Commercial Fats and Oils, Method Cd 8-53.

4.5.2.1.3 Cold test. The cold test shall be determined in accordance with Official and Tentative Methods of the American Oil Chemists Society, Section C, Commercial Fats and Oils, Method Cc 11-53. Remove bottle after 10 hours.

4.5.2.1.4 Fat. Fat shall be determined in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC), Chapter: Spices and Other Condiments: Section: Dressings for Foods; Method: Total Fat.

4.5.2.1.5 Acidity. Acidity shall be determined in accordance with AOAC, Chapter: Spices and Other Condiments; Section: Dressings for Foods; Method: Total Acidity. The acidity shall be based on the concentration of acid in the total moisture content of the product. An example of the calculation for percent acid in the water phase is as follows: If the product contains 16.0 percent of moisture and 0.4 percent of acid, there is a concentration of 0.4 parts of acid in 16.0 parts of water, or a concentration of acid in water phase equal to $\frac{0.4 \times 100}{16}$ or 2.50 percent of acid (calculated as acetic).

4.5.2.1.6 Moisture. Moisture shall be determined in accordance with AOAC Chapter: Spices and Other Condiments; Section: Dressings for Foods; Method: Total Solids.

4.5.2.1.7 Egg yolks. The percentage of total nitrogen and lipid phosphorus (P_2O_5) in the product shall be determined in accordance with AOAC, Chapter; Eggs and Egg Products; Method: Nitrogen; and Method: Lipids and Lipid Phosphorus. Whether liquid or frozen yolks, or liquid or frozen whole eggs are used, testing 43 percent egg yolk solids, the following formula shall apply:

Where: Percent yolk = 94.26 P - 2.192N.
P = Percent of P_2O_5.
N = Percent total nitrogen.

4.5.2.1.8 Sugars. Sugars shall be determined in accordance with AOAC, Chapter: Spices and Other Condiments; Section: Dressings for Foods; Method: Reducing Sugars After Inversion.

4.5.2.1.9 Copper. The copper content shall be determined in accordance with AOAC, Chapter: Metals, Other Elements, and Residues in Foods; Section: Copper; Method: Diethyldithiocarbamate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.5.2.1.10 Linolenic acid. Linolenic acid shall be determined on the oil separated from the product in accordance with Official and Tentative Methods of the American Oil Chemists Society, Section C, Commercial Fats and Oils, Method Cd 7-58. Two standard deviations of 0.25 percent (total 0.5 percent) shall be subtracted from the test result for linolenic acid when the test result is greater than 0.5 percent or 3.0 percent, as applicable (see 3.1.2.). When corn oil is used exclusively as a component in the finished product, two standard deviations of 0.35 percent (total 0.70 percent) shall be used in lieu of 0.25 percent specified above. The actual test result and adjustment shall be reported.

4.5.2.2 Emulsion stability test. The emulsion stability test shall be performed as follows: Place one full primary unopened container in incubator at 100°F (38°C) for 56 hours. There shall be no separation of oil and water phases at the expiration of the 56-hour period in case of types I and II and in 168 hours in case of type IIIA. A slight surface film of oil which may develop during the test may be ignored. Type IIIB is not subject to emulsion stability test.

4.5.2.3 Bacteriological analyses. Bacteriological analyses shall be made in accordance with the following methods given in Compendium of Methods for the Microbiological Examination of Foods.

4.5.2.3.1 Egg products. Egg products shall be tested for standard plate count in accordance with Chapter: Eggs and Egg Products; Section: As applicable for the kind of egg products used.

4.5.2.4 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cans (type I and II product). Type I or type II product, as specified shall be filled in the quantities specified in table VII into metal cans complying with level A requirements of PPP-C-29. In addition to the requirements of PPP-C-29, the metal cans shall have the interior surface double enameled and the interior side seam striped. Prior to closing the can, the headspace shall be gas-flushed with carbon dioxide gas to the extent that the filled and sealed cans shall show a vacuum of not less than 4 inches (87 kPa) of mercury prior to shipment. The vacuum test shall be conducted in accordance with the vacuum test specified in 4.3.4.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE VII. Net weight of primary containers 1/ 2/ 3/

Item	1/2 pint bottle	401 x 411 can	Quart jar	Gallon jar
Type I Mayonnaise		1 lb 8 oz (0.68 kg)	1 lb 14 oz (0.85 kg)	7 lbs 7 oz (3.37 kg)
Type II Salad dressing		1 lb 12 oz (0.79 kg)	2 lbs 1 oz (0.94 kg)	8 lbs 3-1/2 oz (3.72 kg)
Type III French dressing		-	2 lbs (0.91 kg)	8 lbs (3.62 kg)
Type IV Thousand Island dressing	8 oz (0.23 kg)	-	2 lbs (0.91 kg)	8 lbs (3.62 kg)
Type V Blue Cheese dressing	8 oz (0.23 kg)	-	1 lbs 15 oz (0.88 kg)	7 lbs 12 oz (3.52 kg)
Type VI Italian dressing	8 oz (0.23 kg)	-	2 lbs (0.91 kg)	8 lbs (3.62 kg)
Type VII Cole Slaw dressing	8 oz (0.23 kg)	-	2 lbs (0.91 kg)	8 lbs (3.62 kg)
Type VIII French dressing	8 oz (0.23 kg)	-	2 lbs (0.91 kg)	8 lbs (3.62 kg)
Type IX Tartar sauce		-	2 lbs 1 oz (0.94 kg)	8 lbs 3-1/2 oz (3.73 kg)
Type X Green Goddess dressing	8 oz (0.23 kg)	-	2 lbs (0.91 kg)	8 lbs (3.62 kg)

1/ Net weight in any container shall be not more than two percent below the specified net weight.

2/ For 8-ounce bottles, the contents shall be 8 fluid ounces (237 ml).

3/ Metric equivalents of sizes given in this table are: 8 fluid oz - 237 ml; 1 lb 8 oz - 680 grams; 1 lb 12 oz - 794 grams; 1 lb 14 oz - 851 grams; 1 lbs 15 oz - 879 grams; 2 lbs - 907 grams; 2 lbs 1 oz - 936 grams; 7 lbs 7 oz- 3.4 kg; 7 lbs 12 oz - 3.5 kg: 8 lbs - 3.6 kg; 8 lbs 3-1/2 oz - 3.7 kg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.1.1.2 Glass containers. All types of product in amount specified (see 6.1), shall be packaged in 1/2 pint (237 ml) narrow mouth glass bottles or 1-quart (946 ml) or 1 gallon (3.8 liters) glass jars with hermetically sealed closures. Fill shall be in accordance with table VII. The closures may be continuous thread, milled screwcap with curled edge made from commercial 0.25-pound (.113-kg) per base box electrolytic tinplate (see 4.5.2.4) or a hermetic friction-type side seal closure with pry-off feature made from commercial 0.25-pound (.113 kg) per base box electrolytic tinplate. The closures shall be coated outside with a corrosion-resistant coating and inside with a coating suitable for the product. The closures shall be lined with a spot or disc suitable for the product, i.e., pulpboard backing covered with a tin or vinyl spot, enameled hot-dip tinplate, or specially treated aluminum. When screw cap closures are used they shall be secured in place with a shrink-on cellulose band or pressure sensitive, plastic film adhesive tape 3/4-inch (19 mm) wide, applied around the circumference at the juncture of the cap and the bottle. The tape shall overlap itself a minimum of 1 inch (25 mm).

5.1.2 Level C (commercial packaging for all types). The product shall be packaged in accordance with 5.1.2.1 or 5.1.2.2, as specified (see 6.1). The product shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first consuming activity. The supplier may use his standard practice when it meets these requirements.

5.1.2.1 Glass containers. The product shall be packaged in 1/2-pint (237 ml) narrow mouth glass bottles, or 1-quart (946 ml) or 1-gallon (3.8 liters) glass jars, as applicable, to afford adequate protection against deterioration and damage during shipment from the supply source to the first consuming activity. The supplier may use his standard practice when it meets these requirements.

5.1.2.2 Individual servings. One half ounce (14 grams) of the product shall be packaged in boats, cups, or pouches conforming to the requirements of PPP-I-350.

5.2 Packing. The product shall be packed in accordance with level A, B or C, as specified (see 6.1).

5.2.1 Cans, levels A, B or C. The cans of the product shall be packed in accordance with PPP-C-29 for the level specified (see 6.1).

5.2.2 Glass containers, levels A, B or C. The jars or bottles of the product shall be packed in accordance with PPP-G-460 for the level specified (see 6.1).

5.2.3 Individual servings, levels B and C. Individual servings shall be packed in accordance with PPP-I-350 for the level specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.3 Unit loads. When specified (see 6.2), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for type and class of load specified.

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Any commercial labeling, or additional labeling, as specified, that complies with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated There under is acceptable.

5.4.1.2 Shipping containers. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military agencies.

5.4.2.1 Unit containers.

5.4.2.1.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following information shall be included:

REFRIGERATE AFTER OPENING - DO NOT FREEZE

5.4.2.1.2 Glass jars or bottles (for overseas shipment). The label shall be long enough to encircle the jar or bottle with an overlap of at least 1/4 inch (6 mm). Labels for level A glass containers shall be affixed with an adhesive capable of passing the water resistance examination specified in 4.5.1. Labeling shall be in accordance with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated There under.

REFRIGERATE AFTER OPENING - DO NOT FREEZE

(For type III, style B product only)

SHAKE WELL BEFORE USING

5.4.2.1.3 Glass jars or bottles (for domestic shipment). Labeling shall be in accordance with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

5.4.2.2 Shipping containers. Shipping containers for cans and glass containers shall be marked in accordance with MIL-STD-129. The following precautionary markings shall appear prominently on the top of the shipping containers:

STORE IN COOL PLACE (for all classes of products)

DO NOT FREEZE

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:43Z
Check the source to verify that this is the current version before use.

5.4.2.2.1 Individual servings. Intermediate cartons and boxes and shipping containers for individual servings shall be marked in accordance with PPP-I-350

5.4.2.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Type, style, and class of product required (see 1.2)
(c) When a bid sample is required (see 3.1.1).
(d) When finished product is to be shipped in other than ten days after manufacture and packaging (see 3.3.3)
(e) When inspection of plants is required for Civil agencies (see 4.2.1).
(f) Amount of product (all types) to be packaged in glass containers (see 5.1.1.2)
(g) Type of container required for level C packaging (see 5.1.2)
(h) Level of packaging and packing required (see 5.1 and 5.2).
(i) The type and class of unit load when unit loading is required (see 5.3).

6.2 Technical information. Storage temperatures between 40° (4°C) and 60°F (16°C) are recommended. Do not store below 35° (2°C) nor above 90°F (32°C). Freezing temperatures have an undesirable effect on the stability of dressing products. Avoid sudden temperature changes. Keep products away from steam pipes or steam tables and avoid excessive dampness and sunlight. For maximum flavor and freshness, dressing products should be purchased in limited quantities for early use. When held too long they lose freshness, both in taste and color. They should be regarded as semiperishable items even though there is no provision in the procurement system for a semiperishable product. When received in good condition, fresh from production, and properly packaged in cans or jars, dressing products have keeping times of the following approximate lengths (see PPP-I-350 - storage life data for individual servings):

Product	40°F (4°C)		70°F (21°C)		90°F (32°C)
	Class 1	Class 2	Class 1	Class 2	Class 1 & Class 2
	Months		Months		Months
Mayonnaise	12	8	6	4	2
Salad Dressing	12	8	6	4	2
French Dressing	10	6	5	4	2
Thousand Island Dressing	10	6	5	4	2
Blue Cheese Dressing	10	6	5	4	2
Italian Dressing	10	6	5	4	2
Cole Slaw Dressing	10	6	5	4	2
Tartar Sauce	10	6	5	4	2
Green Goddess Dressing	10	6	5	4	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Dressing products packaged in large size containers deteriorate faster than those packaged in smaller containers; this is particularly true when the large size containers are gallon jars.

6.3 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.4 Military agencies. Based on the conditions known or expected to be encountered during shipment, storage, and handling of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

VA-DMS
HEW-FDA
USDA-FSQS
GSA-FSS

Project No. 8950-0254

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.